

BUSINESS REFLECTION QUESTIONS

1. Which factors appear to influence subscription renewal the most?

The factors that appear to influence subscription renewal the most are **monthly spending, watch hours per week, age, number of devices used, and advertisement clicks**. Users who spend more money and watch content more frequently are generally more engaged with the platform and are therefore more likely to renew their subscriptions. The Decision Tree model's feature importance analysis can be used to identify the most influential factors.

2. Why is subscription renewal a classification problem?

Subscription renewal is a classification problem because the target variable has **two possible categories**: **Yes** (the user renews the subscription) and **No** (the user does not renew the subscription). Since the objective is to predict which category a user belongs to, classification algorithms such as Decision Trees and K-Nearest Neighbors are appropriate for this task.

3. Why is monthly spending a regression problem?

Monthly spending is a **regression problem** because the target variable is a **continuous numerical value**. The objective is to predict the exact amount a user is likely to spend each month rather than assigning them to a category. Regression algorithms are designed to estimate such continuous values.

4. Which algorithm performed better for renewal prediction?

Based on the results obtained in this analysis, the **K-Nearest Neighbors (KNN)** model achieved a higher accuracy than the **Decision Tree** model. Therefore, KNN performed better for predicting whether a user would renew their subscription. However, the final conclusion should be based on the actual accuracy values obtained during model evaluation.

Example: If KNN accuracy = 60% and Decision Tree accuracy = 56.67%, then KNN is the better-performing model.

5. How could the platform use these predictions to improve customer retention?

Netflix can use these predictions to **identify users who are at risk of not renewing** their subscriptions. The company can then take proactive measures such as offering **personalized recommendations, providing discounts or promotional offers, sending targeted notifications, and improving user engagement** through customized content suggestions. These actions can help increase customer satisfaction, reduce churn, and improve overall customer retention.

